

Aim

To find the Laplace transform of a unit ramp signal using MATLAB and then plot the signal

Theory

Laplace Transform: The Laplace Transform is an integral transform that converts of time (t) into a function of complex frequency (s). It is used for analyzing time-invariant systems (LTI) which are unstable, or to analyze transient responses or solve differential conditions with initial conditions.

The general formula is given by:

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$

In MATLAB, we will sue the inbuilt *laplace()* function from the functional math toolbox and we will use *syms* to declare symbolic variables.

MATLAB Code

```
Laplace Transform

close all
clearvars

%% To find the Laplace transform of a unit ramp signal

syms s t

% t = 0 : 10^-3 : 50;

t_num = 0:1e-3:50;

x = t;

X = laplace(x, t, s);

y = ilaplace(X, s, t);

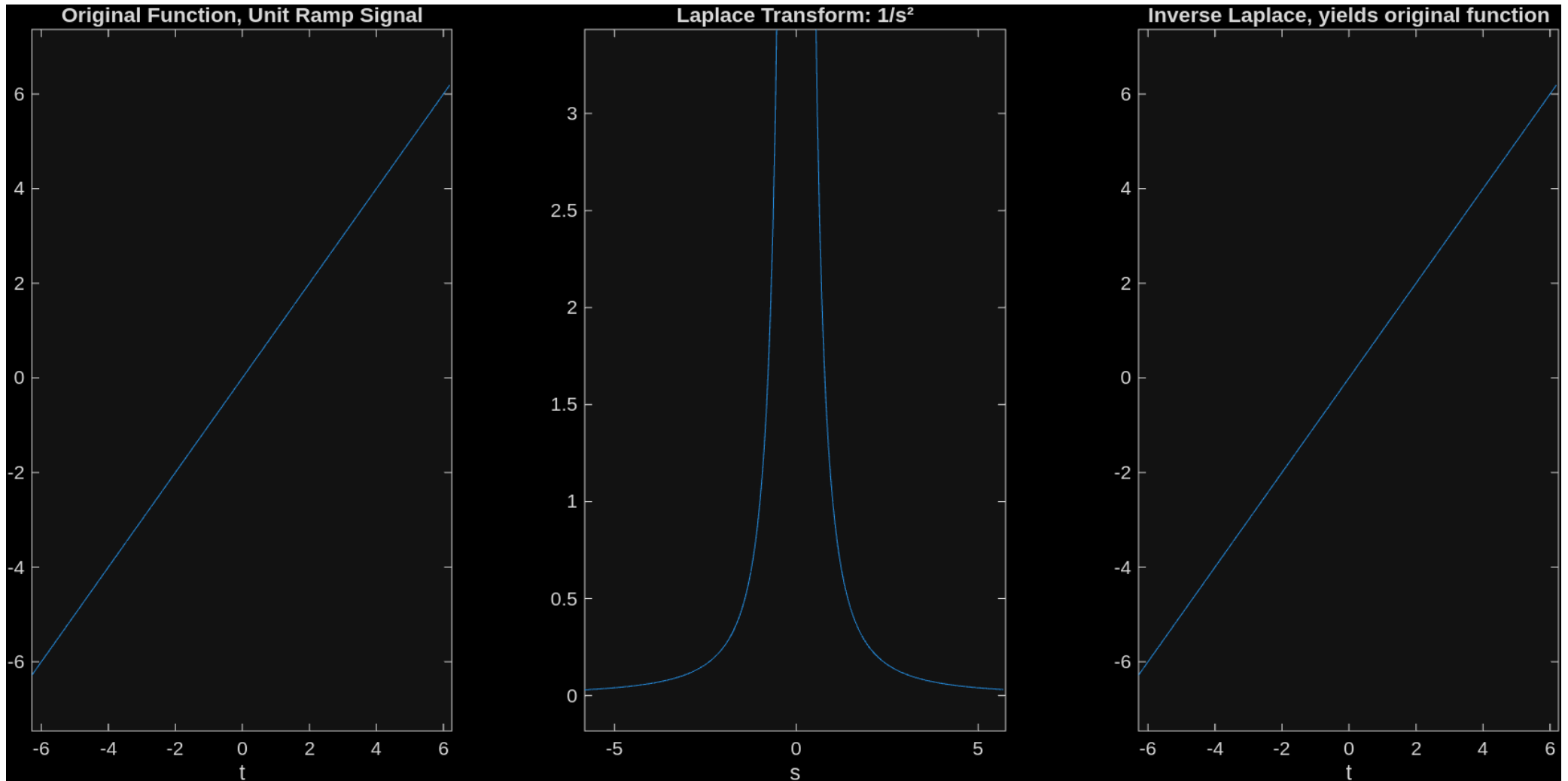
figure;
subplot(1,3,1)
ezplot(x);
title('Original Function, Unit Ramp Signal')

pretty(X);

subplot(1,3,2)
ezplot(X);
title('Laplace Transform: 1/s^2')
pretty(y);

subplot(1,3,3)
ezplot(y);
title('Inverse Laplace, yields original function')
```

Output



Conclusion

We successfully found the Laplace transform of a unit ramp signal, defined as $r(t) = t$, was successfully determined using MATLAB's Symbolic Math Toolbox. The computational result yielded $(1/s^2)$. We also found how the Laplace transform can be reversed to find the original function.